**1** A current-carrying coil produces a magnetic field.

*For Examiner's Use*

It is suggested that the strength $B$ of the magnetic field at the centre of a flat circular coil is inversely proportional to the radius $r$ of the coil.

Design a laboratory experiment that uses a Hall probe to test the relationship between $B$ and $r$. You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

**(a)** the procedure to be followed,

**(b)** the measurements to be taken,

**(c)** the control of variables,

**(d)** the analysis of the data,

**(e)** the safety precautions to be taken.

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